SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
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COMPUTER SCIENCE AND ENGINEERING

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ASSIGNMENT – 2

Name : T Raju

Reg_No : 192465038

Course Code : **CSA63**

Course Name : **THREAT INTELLIGENCE AND NETWORK SECURITY**

Max Marks**: 50**

Answer all the Questions

1	The finance office of a university receives an e-mail that appears to come from the vice-chancellor's office, asking the accounts officer to open an attached "budget revision" document and to approve an urgent payment. The attachment contains a macro that installs a trojan, and within hours the finance file server is encrypted. Apply your knowledge of malware and phishing to this case. Explain the types of malware (virus, worm, trojan, ransomware, spyware, rootkit and botnet agent) with their method of spreading, and identify which types are involved here. Explain the types of phishing (bulk, spear phishing, whaling, smishing, vishing and clone phishing) and classify this attack with reasons. Prepare a table of the warning signs in the e-mail (sender address, header mismatch, urgency, link and attachment) and describe the technical controls (SPF, DKIM, DMARC, attachment sandboxing, macro blocking) and the awareness measures that would have stopped the attack.	(10)	(CO2)	(BL3)
2	An online ticket-booking website sells tickets for a popular cricket match. Within seconds of the sale opening, its web servers become unreachable. The monitoring graph shows a very large rise in incoming traffic, thousands of half-open TCP connections, and many repeated requests for the search page from addresses all over the world. Analyse the incident by explaining how DoS and DDoS attacks differ, and the working of volumetric attacks (UDP flood and DNS or NTP amplification), protocol attacks (SYN flood) and application-layer attacks (HTTP flood), with the structure of the botnet that launches them. Explain how the security team can distinguish a DDoS attack from a genuine rush of customers. Recommend a response and prevention plan using rate limiting, SYN cookies, content delivery networks, traffic scrubbing, anycast and black-holing, and discuss the cost and side effects of each measure.	(10)	(CO2)	(BL4)

3	Sales staff of a company often work from a café that offers free public Wi-Fi. A security audit finds that an attacker sitting in the café could create a fake access point with the same name as the café network, redirect users to a copy of the company's web mail login page, and capture the data that passes over the network. Analyse the attacks that make this possible by explaining the working of man-inthe-middle attacks, the evil-twin access point, ARP spoofing, DNS spoofing, IP and	(10)	(CO2)	(BL4)
	e-mail spoofing, SSL stripping and packet sniffing, with a labelled sequence or network diagram of one attack. Differentiate passive from active attacks in this case. Describe how each attack is detected (for example, duplicate MAC addresses in the ARP table, certificate warnings and IDS alerts) and prevented (VPN, HTTPS with HSTS, WPA3, dynamic ARP inspection, port security and 802.1X), and discuss which measures the company can apply to staff laptops it does not control.			
4	A small private clinic has a reception computer, two doctors' laptops, a patientrecords server, a network printer, a Wi-Fi router and a few IP cameras on one flat network. The clinic has never tested its network for weaknesses. Apply the vulnerability assessment process to this network. Explain the steps of asset discovery, scanning (for example, with Nmap, Nessus or OpenVAS), identification of vulnerabilities, scoring with CVSS, prioritisation, reporting, remediation and rescanning; the difference between authenticated and unauthenticated scans and between network-based and host-based scans; and the difference between a vulnerability assessment and a penetration test. Prepare a sample findings table with at least five realistic vulnerabilities of this clinic (for example, default router password, outdated operating system, unpatched server software, open remote desktop port and unencrypted camera streams), giving for each the affected asset, CVSS severity rating, possible attack and recommended remediation.	(10)	(CO2)	(BL3)
5	The Security Operations Centre of a bank receives threat data from three kinds of sources: its own firewall, proxy and e-mail logs; free OSINT sources such as community threat-sharing platforms, malware-analysis sites, vulnerability databases and internet-scanning services; and a paid commercial feed. A new batch contains these entries: an IP address that was reported as malicious fourteen months ago, a domain that imitates the bank's login page and was reported yesterday, and a file hash that also matches a legitimate Windows utility. Evaluate the threat data sources and the quality of these entries. Compare internal, OSINT, community and commercial sources, explain the types of Indicators of Compromise (hashes, IP addresses, domains, URLs, e-mail artefacts, registry keys and file names) with the Pyramid of Pain, and judge each entry on timeliness, accuracy, relevance, context and the risk of false positives. Explain how feed formats such as STIX and TAXII help sharing, and recommend how the bank should select, integrate and retire indicators in its SIEM, including aging of indicators and removal of duplicates.	(10)	(CO2)	(BL5)

QUESTION 1: MALWARE AND PHISHING ATTACKS

Introduction

Malware is malicious software designed to damage computers, steal information or gain unauthorised access. Phishing is a cyberattack in which attackers trick users into revealing sensitive information or downloading malicious files. In this case, an attacker sends an email pretending to be from the Vice-Chancellor's office. The accounts officer opens an infected budget document, which installs a trojan. Later, ransomware encrypts the university's finance server. **A. Types of Malware**

Malware type	Explanation	Method of spreading
Virus	Malicious code that attaches itself to files or programs.	Spreads when infected files are shared or executed.
Worm	Malware that spreads automatically without user interaction.	Spreads through network vulnerabilities.
Trojan	Malware that looks like a genuine or useful application.	Spreads through fake software, email attachments and downloads.
Ransomware	Malware that locks or encrypts files and demands money.	Spreads through phishing emails, malicious downloads and vulnerabilities.
Spyware	Software that secretly monitors users and steals information.	Spreads through infected applications and websites.
Rootkit	Malware that hides its presence and gives attackers privileged access.	Spreads through infected software or system vulnerabilities.
Botnet agent	Malware that makes a computer part of an attacker-controlled network.	Spreads through infected devices and security weaknesses.

Malware involved in this case

- Trojan: The infected budget document installs malicious software.
- Ransomware: It encrypts the university's finance files.
- Spyware may also be involved if the attacker steals financial information. **B. Types of Phishing**

Type	Explanation
Bulk phishing	Sending fake emails to a large number of people.
Spear phishing	Sending targeted emails to a specific person or organisation.
Whaling	Targeting senior executives or important officials.
Smishing	Phishing through SMS messages.
Vishing	Phishing through phone calls or voice messages.
Clone phishing	Copying a genuine email and replacing its attachment or link with a malicious one.

Classification: This is mainly spear phishing because the attacker targets the university's accounts officer with a message related to financial responsibilities. It also uses impersonation of a senior official, resembling a whaling-style identity lure.

C. Warning Signs in the Email

Warning sign	Explanation
Sender address	The email address may contain spelling mistakes or use a fake domain.
Header mismatch	The displayed sender name may not match the actual sending address or authentication results.
Urgency	The email demands immediate payment approval to pressure the employee.
Link	Suspicious links may redirect users to fake login pages.

Attachment	The budget document contains a malicious macro.

D. Technical Controls

1. SPF (Sender Policy Framework): Checks whether the sending mail server is authorised to send emails for the domain.
2. DKIM (DomainKeys Identified Mail): Uses a digital signature to help verify that an email is associated with the signing domain and has not been modified.
3. DMARC: Uses SPF and DKIM results to help prevent domain spoofing and tells receiving servers how to handle failed messages.
4. Attachment sandboxing: Opens suspicious attachments in a safe, isolated environment to detect malicious behaviour.
5. Macro blocking: Blocks macros in files received from untrusted sources.

E. Awareness Measures

- Train employees to identify suspicious emails.
- Never open unexpected attachments.
- Verify urgent payment requests through a separate communication channel.
- Check the sender's complete email address.
- Report suspicious emails to the IT security team.
- Use two-person approval for large financial transactions.

Conclusion: The attack could have been prevented through email authentication, macro blocking, attachment scanning and employee awareness training.

Introduction

QUESTION 2: DoS AND DDoS ATTACKS

A Denial of Service (DoS) attack attempts to make a website or server unavailable by overwhelming it with traffic or requests. A Distributed Denial of Service (DDoS) attack performs the same activity using multiple compromised devices from different locations. In this case, the cricket ticket-booking website becomes unavailable because of a sudden increase in malicious network traffic.

A. Difference Between DoS and DDoS

DoS	DDoS
Attack comes from a single source.	Attack comes from multiple sources.
Usually easier to identify.	More difficult to identify and filter.
Generally has limited attack capacity.	Can generate very large traffic volumes.
Blocking the source may stop the attack.	Blocking individual sources is difficult.
Uses one attacking system.	Uses multiple compromised systems.

B. Types of DDoS Attacks

1. Volumetric attacks

These attacks consume network bandwidth by sending a huge amount of traffic.

- UDP flood: Sends large numbers of UDP packets to overwhelm the target network or server.
- DNS amplification: Misuses DNS servers to generate large responses towards a victim, amplifying the traffic. ● NTP amplification: Misuses Network Time Protocol servers to generate amplified traffic towards the victim.

2. Protocol attacks

These attacks target network protocols and consume network or server resources.

- SYN flood: Sends many TCP SYN requests without completing the TCP three-way handshake.
- This creates many half-open connections and can exhaust the server's connection resources.

3. Application-layer attacks

These attacks target specific application services.

- HTTP flood: Sends a large number of HTTP requests to a website.
- Repeated search-page requests consume server processing power and may prevent genuine customers from accessing the website.

C. Structure of a DDoS Botnet

A botnet is a group of infected computers or devices controlled by an attacker. The basic structure is:

Attacker → Command-and-Control (C2) Server → Infected Bots / Devices → Victim Website

6. The attacker controls a command-and-control (C2) server.
7. The C2 server sends instructions to infected devices.
8. The infected devices are called bots or zombies.
9. The bots send traffic towards the victim website.
10. The victim server becomes overloaded and unavailable.

D. Identifying DDoS Attacks vs Genuine Customer Traffic

Genuine traffic	Possible DDoS traffic
Users show normal browsing patterns.	Requests may be repeated continuously.
Traffic usually increases naturally.	Traffic may increase suddenly and sharply.
Requests follow normal user behaviour.	Requests may have abnormal or automated patterns.
Most connections are completed normally.	Many half-open connections may appear.
Traffic may come from real customers.	Traffic may come from suspicious distributed sources.

The security team should examine traffic volume, IP reputation, request frequency, connection completion rates, user behaviour and whether requests are spread across many networks.

E. Response and Prevention Measures

Security measure	Purpose	Cost and side effects
Rate limiting	Limits the number of requests from a source or user.	May block genuine users who share the same IP address.

SYN cookies	Reduces resource use for half-open TCP connections.	May add processing overhead and does not stop every DDoS type.
CDN	Distributes content across multiple servers and absorbs some traffic.	Requires subscription and correct configuration.
Traffic scrubbing	Filters malicious traffic before it reaches the server.	Can be expensive and may introduce latency.
Anycast	Distributes traffic across multiple network locations.	Requires suitable network infrastructure and provider support.
Black-holing	Drops traffic to a targeted IP address or network.	Can also make the legitimate website completely unreachable.

F. Response Plan

11. Identify abnormal traffic using monitoring tools.
12. Activate the DDoS protection service and contact the network or hosting provider.
13. Apply rate limiting and SYN cookies where appropriate.
14. Use a CDN and traffic scrubbing to filter malicious traffic.
15. Monitor network traffic and application performance continuously.
16. Restore normal services after the attack.
17. Prepare an incident report and improve security controls.

Conclusion: A combination of CDN, rate limiting, SYN cookies and traffic scrubbing can reduce the impact of DDoS attacks and improve website availability.

Introduction

QUESTION 3: MAN-IN-THE-MIDDLE (MITM) ATTACKS

A Man-in-the-Middle (MITM) attack occurs when an attacker secretly places themselves between two communicating devices and intercepts or modifies the information being exchanged. In this case, company employees use free public Wi-Fi at a café. An attacker creates a fake Wi-Fi access point and redirects employees to a fake company email login page to steal their credentials.

A. Types of Attacks

1. Man-in-the-Middle attack

The attacker secretly intercepts communication between two users or devices. The attacker may read or modify transmitted information. Example: stealing login credentials through public Wi-Fi.

2. Evil Twin Access Point

An attacker creates a fake Wi-Fi network with the same name as a genuine network. Users mistakenly connect to the fake network, allowing the attacker to monitor or manipulate traffic.

3. ARP Spoofing

ARP is used to identify the MAC address associated with an IP address in a local network. An attacker sends fake ARP messages so that the victim may send network traffic through the attacker's device.

4. DNS Spoofing

The attacker provides a fake IP address for a genuine website. Users may be redirected to a malicious website, such as a fake company email login page.

5. IP and Email Spoofing

- IP spoofing: The attacker changes the source IP address of network packets to make them appear to come from another system.

Attack	Detection method
MITM	Monitor unusual network connections and certificate warnings.
Evil Twin	Check suspicious Wi-Fi names and unexpected access points.
ARP spoofing	Detect duplicate or changing MAC addresses in the ARP table.
DNS spoofing	Check unexpected DNS responses and incorrect IP addresses.

IP spoofing	Monitor suspicious source IP addresses using firewalls and

- Email spoofing: The attacker forges the sender address to make an email appear to come from a trusted person or organisation.

6. SSL Stripping

The attacker attempts to downgrade a secure HTTPS connection to an insecure HTTP connection. This may expose information if the victim accepts the insecure connection. HTTPS with HSTS helps prevent this attack.

7. Packet Sniffing

Packet sniffing means capturing and analysing network traffic. Attackers may use it to observe transmitted information. For example, they may capture unencrypted login information.

B. Labelled Network Diagram and Attack Sequence

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EMPLOYEE LAPTOP\n      | connects to fake café Wi-Fi\n      ▼\nATTACKER / EVIL TWIN ACCESS POINT\n      | intercepts or redirects traffic\n      ▼\nFAKE COMPANY WEBMAIL LOGIN PAGE\n      | employee enters credentials\n      ▼\nATTACKER CAPTURES CREDENTIALS
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18. The employee connects to the attacker's fake café Wi-Fi.
19. The employee opens the company webmail website.
20. The attacker intercepts or redirects the communication.
21. The employee sees a fake login page.
22. The employee enters the username and password.
23. The attacker captures the credentials.

C. Passive and Active Attacks

Passive attack	Active attack
Monitors communication without changing it.	Modifies or disrupts communication.
Mainly used to collect information.	Used to manipulate information or redirect users.
Often difficult to detect.	May be easier to detect through unusual network behaviour.
Example: packet sniffing.	Examples: ARP spoofing and DNS spoofing.

D. Detection Methods

	IDS.
Email spoofing	Check sender details and email authentication results.
SSL stripping	Detect HTTPS downgrade attempts and certificate warnings.
Packet sniffing	Monitor suspicious network interfaces and IDS alerts.

E. Prevention Methods

24. VPN: Encrypts network communication between the employee's device and the VPN server.
25. HTTPS with HSTS: Enforces secure HTTPS connections and helps prevent HTTP downgrades for protected websites.
26. WPA3: Provides stronger wireless network security and encryption.
27. Dynamic ARP Inspection: Checks ARP messages to help prevent ARP spoofing on supported managed switches.
28. Port security: Restricts unauthorised devices on network switch ports.
29. 802.1X authentication: Allows only authenticated users or devices to access supported networks.

F. Security Measures for Staff Laptops ●

Install endpoint security software.

- Enforce VPN usage while accessing company resources.
- Enable automatic operating system and application updates.
- Disable automatic connection to unknown Wi-Fi networks.
- Use HTTPS websites with HSTS.
- Enable multi-factor authentication (MFA).
- Apply device-management policies where supported.

Conclusion: MITM attacks can steal confidential information through insecure networks. Using VPN, HTTPS, MFA and secure laptop configurations can significantly reduce these risks.

Introduction

QUESTION 4: VULNERABILITY ASSESSMENT

A vulnerability assessment is a systematic process of identifying, analysing and evaluating security weaknesses in a computer network or system. In this case, the private clinic has reception computers, doctors' laptops, a patient-records server, a printer, a Wi-Fi router and IP cameras connected to one network. Because patient information is confidential, the clinic must identify and fix security weaknesses.

A. Steps in Vulnerability Assessment

Asset discovery: Identify all devices connected to the clinic network. Record IP addresses, MAC addresses, operating systems and device types. Examples include the reception computer, doctors' laptops, server, printer and cameras.
Scanning: Use security tools to identify open ports, running services and known weaknesses. Nmap discovers devices, open ports and services; Nessus and OpenVAS identify known vulnerabilities and misconfigurations.
Identification of vulnerabilities: Analyse scanning results and identify outdated software, weak passwords, unnecessary open ports and missing security updates.
CVSS scoring: Use the Common Vulnerability Scoring System to measure the severity of a vulnerability.
Prioritisation: Fix critical and high-severity vulnerabilities first, especially those affecting patient records. Consider exploitability and business impact.
Reporting: Prepare a report containing vulnerabilities, affected devices, severity ratings, possible impacts and recommended solutions.
Remediation: Fix weaknesses by updating software, changing default passwords, closing unnecessary ports and applying secure configurations.
Re-scanning: Scan again after remediation to verify that the vulnerabilities have been fixed. **CVSS Severity Ratings**

CVSS score	Severity
0.1–3.9	Low
4.0–6.9	Medium
7.0–8.9	High
9.0–10.0	Critical

B. Authenticated and Unauthenticated Scanning

Authenticated scanning	Unauthenticated scanning
Uses valid login credentials.	Does not require login credentials.
Examines internal system configurations.	Examines externally visible weaknesses.
Provides more detailed results.	Provides more limited information.
Can identify missing patches and weak configurations.	Can identify open ports and exposed services.

C. Network-Based and Host-Based Scanning

Network-based scanning	Host-based scanning
Scans devices across a network.	Scans individual computers or servers.
Identifies open ports and network services.	Identifies missing patches and local weaknesses.
Examines network-level security.	Examines system-level security.
Example: Nmap network discovery.	Example: a local vulnerability assessment agent.

D. Vulnerability Assessment vs Penetration Testing

Vulnerability assessment	Penetration testing
Identifies security weaknesses.	Tests whether weaknesses can be exploited.
Focuses on finding vulnerabilities.	Focuses on demonstrating security risks.
Usually uses automated scanning tools.	Uses manual and automated testing.
Provides a list of vulnerabilities.	Provides evidence of successful exploitation where authorised.
Helps improve overall security.	Helps understand the actual impact of attacks.

E. Sample Vulnerability Findings Table

The following are realistic example findings. CVSS scores are illustrative estimates; actual scores depend on the exact product, version and configuration.

Affected asset	Vulnerability	CVSS	Severity	Recommended

				remediation
Wi-Fi router	Default administrator password	9.8	Critical	Change the default password and enable MFA if supported.
Reception PC	Outdated operating	8.1	High	Update the operating
	system			system and install security patches.
Patient records server	Unpatched server software	9.1	Critical	Install security updates and monitor the server.
Doctors' laptop	Unnecessary or exposed Remote Desktop port	8.1	High	Restrict RDP access and use VPN with MFA.
IP cameras	Unencrypted camera streams	7.5	High	Enable encrypted streaming and change default credentials.
Network printer	Unnecessary open services	5.3	Medium	Disable unused services and restrict access.

F. Importance of Vulnerability Assessment

- Protects confidential patient records.
- Prevents unauthorised access.
- Identifies outdated software and weak configurations. ● Reduces the risk of malware and ransomware attacks.
- Improves the overall security of the clinic network.

Conclusion: Regular vulnerability assessments help the clinic discover and fix security weaknesses before attackers can exploit them. Re-scanning ensures that the security improvements are effective.

QUESTION 5: THREAT INTELLIGENCE AND INDICATORS OF COMPROMISE (IOC)

Introduction

Threat Intelligence is the process of collecting, analysing and sharing information about cyber threats to protect computer systems and networks. An Indicator of Compromise (IOC) is a piece of information that indicates a system may have been attacked or compromised. In this case, a bank collects threat information from internal logs, free OSINT sources and paid commercial feeds to identify malicious IP addresses, domains and file hashes. **A. Types of Threat Intelligence Sources**

1. Internal sources

Collect information from the organisation's own security systems. Examples include firewall logs, proxy logs and email security logs. They help identify suspicious activities within the bank.

2. OSINT sources

OSINT stands for Open Source Intelligence. It collects publicly available threat information. Examples include malware-analysis websites, vulnerability databases and internet-scanning services. It helps organisations identify known cyber threats.

3. Community sources

Cybersecurity researchers and security communities share malicious IP addresses, domains and malware details. The quality of the information may vary.

4. Commercial sources

Paid providers offer professionally analysed threat intelligence, updated threat feeds, threat reports and attacker information. These services can be useful but require a subscription.

B. Comparison of Threat Intelligence Sources

Source	Advantages	Limitations
Internal	Organisation-specific information.	Limited to collected internal data.
OSINT	Free and easily accessible.	Information may be outdated or inaccurate.
Community	Shared knowledge and quick updates.	Quality may vary.
Commercial	Professional analysis and regular updates.	Requires subscription and cost.

C. Types of Indicators of Compromise (IOC)

IOC type	Explanation	Example
File hashes	Digital values used to identify files.	SHA-256 hash of a suspicious file.
IP addresses	Addresses associated with suspicious network activity.	Known malicious IP address.
Domains	Website names used by attackers.	Fake bank login domain.
URLs	Links that direct users to malicious websites.	Fake login URL.
Email artefacts	Information used to identify suspicious emails.	Forged sender address or message ID.
Registry keys	Windows registry entries modified by malware.	Suspicious startup entry.
File names	Names of suspicious or malicious files.	Unknown executable file.

D. Pyramid of Pain

The Pyramid of Pain explains how difficult it is for attackers to change different indicators after defenders identify them.

Level	Indicator	Difficulty for attacker to change
1	Hash values	Very easy
2	IP addresses	Easy
3	Domain names	Simple
4	Network / host artefacts	Annoying
5	Tools	Challenging

6	Tactics, Techniques and Procedures (TTPs)	Tough

Hash values are easy for attackers to change by modifying a file. IP addresses and domains can also be replaced. Network and host artefacts, tools, and especially TTPs require more effort to change. Therefore, TTP-based intelligence is generally more difficult for attackers to avoid than simple hash-based detection.

E. Evaluation of the Given Threat Data

1. Malicious IP address reported fourteen months ago ●

Timeliness: Low because the information is old.

- Accuracy: It may be inaccurate if the IP address is no longer malicious.
- Relevance: Needs further investigation.
- False-positive risk: High because IP addresses may be reassigned to legitimate users.

2. Domain imitating the bank's login page reported yesterday ●

Timeliness: High because it was reported recently.

- Accuracy: High if the report is verified.
- Relevance: Very high because it targets the bank.
- False-positive risk: Low if the domain is confirmed as fraudulent.

3. File hash matching a legitimate Windows utility

- Timeliness: Depends on when the hash was collected.
- Accuracy: Uncertain because the hash also matches a legitimate utility.
- Relevance: Requires further investigation.
- False-positive risk: High if the hash is used without checking file context and authenticity. **F. Comparison of the Three Indicators**

Indicator	Timeliness	Relevance	False-positive risk	Recommended action
Old malicious IP	Low	Medium	High	Recheck before blocking.
Fake bank domain	High	Very high	Low if verified	Block and monitor after verification.
Windows utility hash	Uncertain	Medium	High	Investigate file context and authenticity before action.

G. STIX and TAXII

1. STIX (Structured Threat Information Expression)

- STIX is a standard language used to represent and share cyber threat intelligence.
- It organises information about attackers, malware, indicators and cyber incidents.
- It helps different security tools understand threat information in a common format.

2. TAXII (Trusted Automated Exchange of Intelligence Information)

- TAXII is a protocol used to exchange threat intelligence between organisations and security systems.
- It supports automated sharing of threat information.
- It helps security teams receive updated threat feeds.

STIX	TAXII
Describes threat intelligence.	Transfers threat intelligence.
Provides a common data format.	Provides a communication mechanism.
Organises threat information.	Shares threat information between systems.

H. Integrating Threat Intelligence into SIEM

SIEM stands for Security Information and Event Management. It collects and analyses security logs from different systems to detect suspicious activities.

30. Select indicators: Collect relevant IOCs from internal, OSINT, community and commercial sources.
31. Verify indicators: Check whether the information is accurate and trustworthy.

32. Integrate feeds: Import threat intelligence into the SIEM using supported formats such as STIX and TAXII.
33. Detect threats: Compare network and system logs against known malicious indicators.
34. Remove duplicates: Avoid storing the same IP addresses, domains or hashes multiple times.
35. Age indicators: Reduce the priority of old indicators and periodically review their validity.
36. Retire indicators: Remove expired or confirmed false-positive indicators from active detection rules.

I. Recommendations

- Use multiple trusted threat intelligence sources.
- Prioritise recent and relevant indicators.
- Verify suspicious IP addresses and domains before blocking them.
- Avoid blocking file hashes without proper investigation.
- Update threat feeds regularly.
- Integrate threat intelligence with SIEM.
- Remove outdated and duplicate indicators.
- Monitor security alerts continuously.

Conclusion: Threat intelligence helps banks identify and respond to cyber threats before they cause serious damage. Using reliable sources, verified IOCs, STIX, TAXII and SIEM integration improves threat detection and overall security.